

Random Matrices

M1R Project

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Overview

I Central Limit Theorem: Before digging in to random matrices we discuss a baby-version of universality coming from adding many random variables, using the Method of Moments to relate to the number of partitions of a set into pairs.

II Universality: Many families of random matrices have the same eigenvalue statistics which can be explained by adapting the Method of Moments to relate to the number of *non-crossing* partitions of a set into pairs.

III Free Probability: Adding or multiplying random matrices satisfy special rules of (non-commutative) free random variables. These can be explained again by the Method of Moments.

IV Randomised Linear Algebra: We finish with a nice application of using randomness to reduce linear algebra problems like linear regression to lower dimensional versions which are faster to solve.



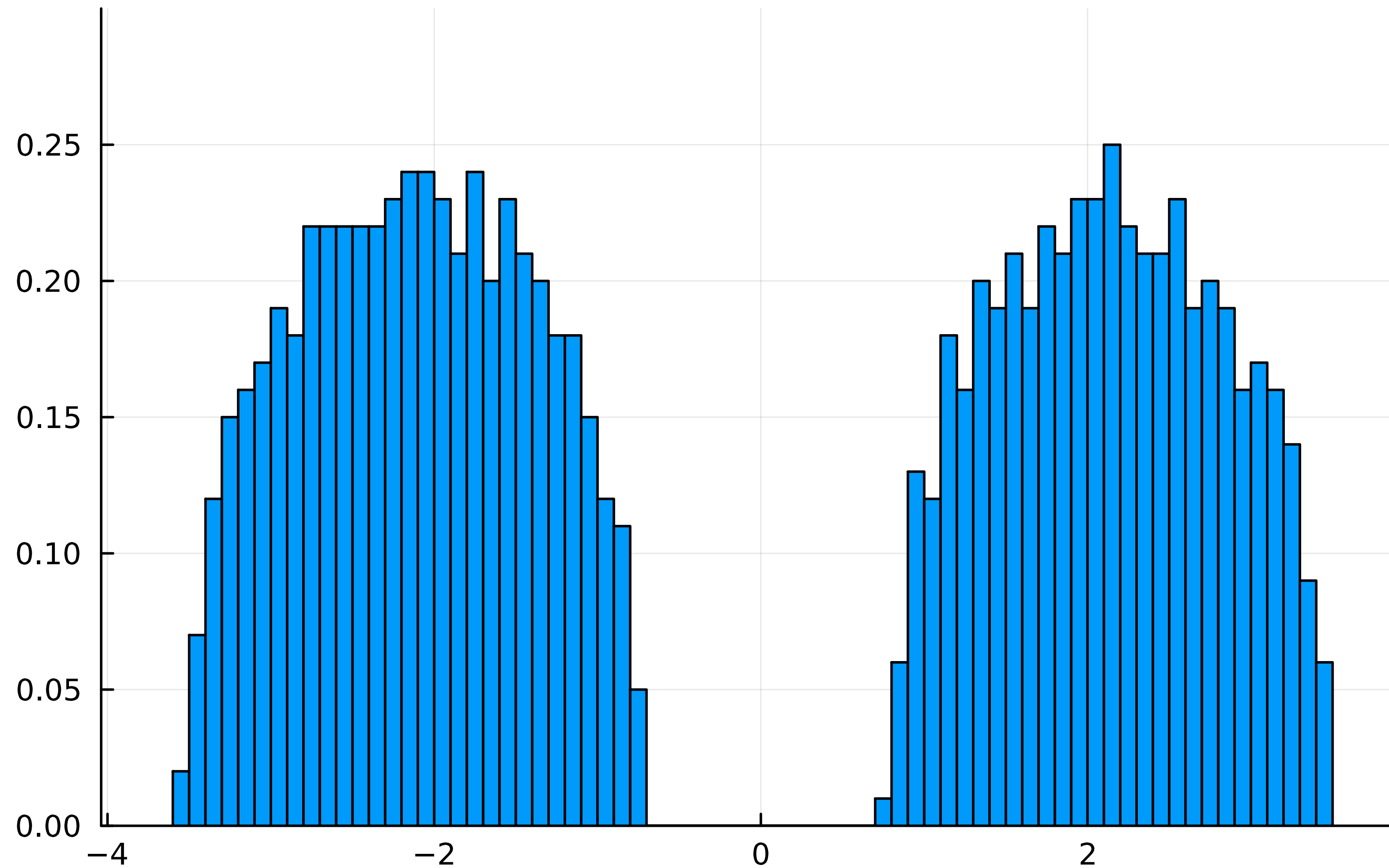
Link to notes

III Free Probability

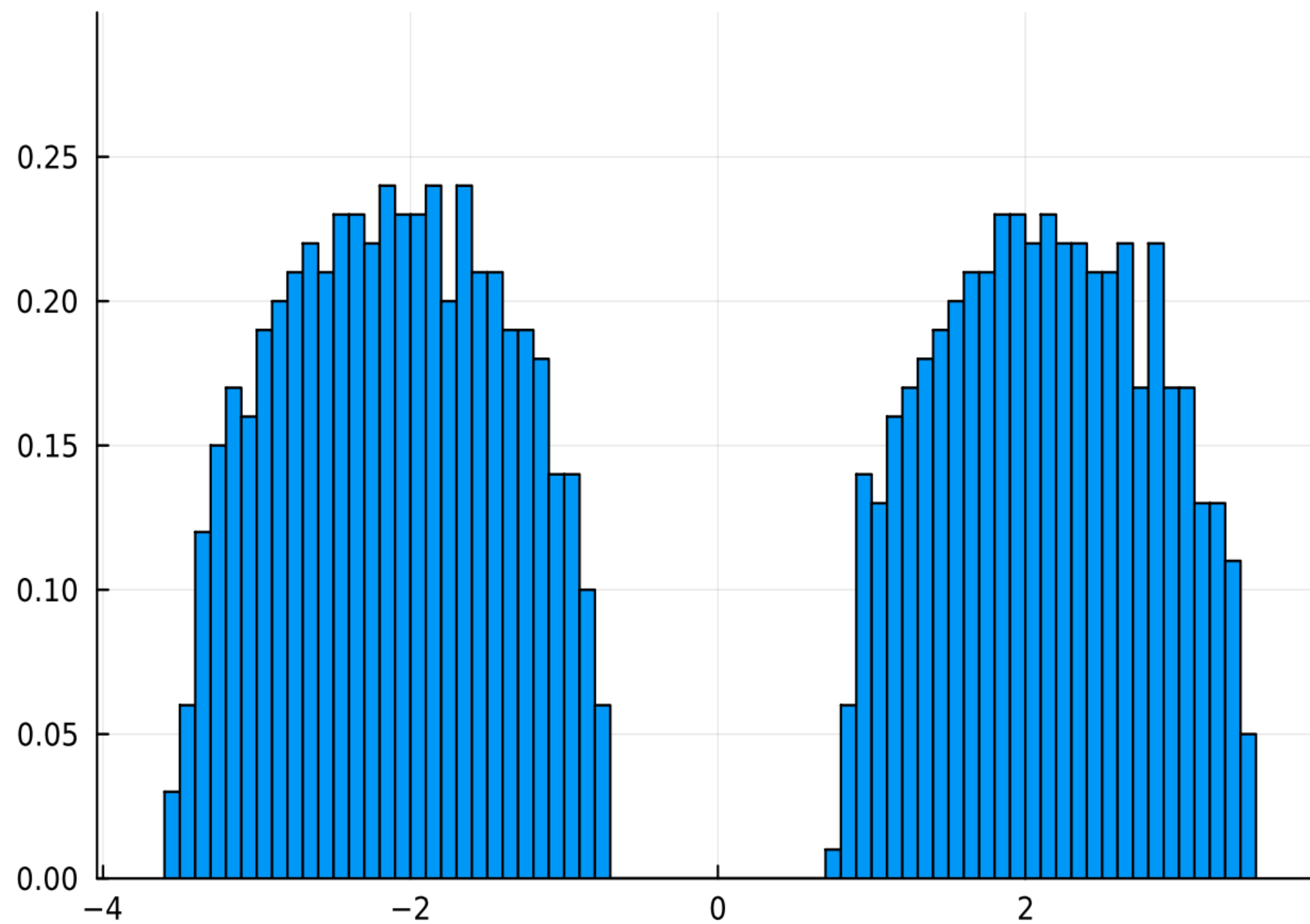
III.1 Experimental observation of free convolutions

Eigenvalues of addition of random matrices does not depend on eigenvectors

Eigenvalues of $A_n + D_n$



Eigenvalues of $A_n + B_n$



III.2 Free random variables

A non-commutative version of probability

Definition 4 (free random variables). A and B are *free random variables* equipped with an expectation $\mathbf{E}$ if for all polynomials $f_1, \dots, f_n$ and $g_1, \dots, g_n$ such that $\mathbf{E}f_k(A) = \mathbf{E}g_k(B) = 0$ we have

$$\mathbf{E}[f_1(A)g_1(B) \dots f_n(A)g_n(B)] = 0,$$

where the identity element satisfies $\mathbf{E}1 = 1$.

$$\begin{aligned}
0 &= \mathbf{E}[(A - \mathbf{E}A)(B - \mathbf{E}B)(A - \mathbf{E}A)(B - \mathbf{E}B)] \\
&= \mathbf{E}[ABAB] - \underbrace{\mathbf{E}[ABA] \mathbf{E}B}_{=\mathbf{E}A^2\mathbf{E}B} - \underbrace{\mathbf{E}A \mathbf{E}[AB^2]}_{=\mathbf{E}A\mathbf{E}B^2} + \underbrace{\mathbf{E}[AB] \mathbf{E}A\mathbf{E}B}_{=\mathbf{E}A\mathbf{E}B} \\
&\quad - \underbrace{\mathbf{E}[A^2B] \mathbf{E}B}_{=\mathbf{E}A^2\mathbf{E}B} + \mathbf{E}A^2\mathbf{E}[B]^2 + \underbrace{\mathbf{E}[AB] \mathbf{E}A\mathbf{E}B}_{=\mathbf{E}A\mathbf{E}[B]} - \mathbf{E}[A]^2\mathbf{E}[B]^2 \\
&\quad - \mathbf{E}[A] \underbrace{\mathbf{E}[BAB]}_{=\mathbf{E}A\mathbf{E}B^2} + \mathbf{E}A\mathbf{E}B \underbrace{\mathbf{E}[BA]}_{=\mathbf{E}A\mathbf{E}B} + \mathbf{E}[A]^2\mathbf{E}B^2 - \mathbf{E}[A]^2\mathbf{E}[B]^2 \\
&\quad + \mathbf{E}A\mathbf{E}B \underbrace{\mathbf{E}[AB]}_{=\mathbf{E}A\mathbf{E}B} - \mathbf{E}[A]^2\mathbf{E}[B]^2 - \mathbf{E}[A]^2\mathbf{E}[B]^2 + \mathbf{E}[A]^2\mathbf{E}[B]^2 \\
&= \mathbf{E}[ABAB] - \mathbf{E}A^2\mathbf{E}[B]^2 - \mathbf{E}[A]^2\mathbf{E}B^2 + \mathbf{E}[A]^2\mathbf{E}[B]^2 \quad \Rightarrow \\
\mathbf{E}[ABAB] &= \mathbf{E}A^2\mathbf{E}[B]^2 + \mathbf{E}[A]^2\mathbf{E}B^2 - \mathbf{E}[A]^2\mathbf{E}[B]^2
\end{aligned}$$

Lemma 6 (mixed moments). *For $k_1, \dots, k_n$ and $j_1, \dots, j_n$ the mixed moments*

$$\mathbf{E}[A^{k_1} B^{j_1} \dots A^{k_n} B^{j_n}]$$

can be expressed as functions of the moments

$$\mathbf{E}A, \mathbf{E}A^2 \dots, \mathbf{E}A^{k_1 + \dots + k_n}, \mathbf{E}B, \mathbf{E}B^2 \dots, \mathbf{E}B^{j_1 + \dots + j_n}.$$

Lemma 7 (free cyclic property). *Mixed moments have a cyclic property:*

$$\mathbf{E}[A^{k_1} B^{j_1} \dots A^{k_n} B^{j_n}] = \mathbf{E}[B^{j_1} \dots A^{k_n} B^{j_n} A^{k_1}] = \mathbf{E}[B^{j_n} A^{k_1} B^{j_1} \dots A^{k_n}].$$

III.3 Free convolution

Addition of free random variables

$$\begin{aligned}
\mathbf{E}[(A+B)^4] &= \mathbf{E}A^4 + \underbrace{\mathbf{E}[A^3B]}_{=\mathbf{E}A^3\mathbf{E}B} + \underbrace{\mathbf{E}[A^2BA]}_{=\mathbf{E}A^3\mathbf{E}B} + \underbrace{\mathbf{E}[A^2B^2]}_{=\mathbf{E}A^2\mathbf{E}B^2} + \underbrace{\mathbf{E}[ABA^2]}_{=\mathbf{E}A^3\mathbf{E}B} \\
&\quad + \underbrace{\mathbf{E}[ABAB]}_{=\mathbf{E}A^2\mathbf{E}[B]^2 + \mathbf{E}[A]^2\mathbf{E}B^2 - \mathbf{E}[A]^2\mathbf{E}[B]^2} + \underbrace{\mathbf{E}[AB^2A]}_{=\mathbf{E}A^2\mathbf{E}B^2} \\
&\quad + \underbrace{\mathbf{E}[AB^3]}_{=\mathbf{E}A\mathbf{E}B^3} + \underbrace{\mathbf{E}[BA^3]}_{=\mathbf{E}A^3\mathbf{E}B} + \underbrace{\mathbf{E}[BA^2B]}_{=\mathbf{E}A^2\mathbf{E}B^2} + \underbrace{\mathbf{E}[BAB A]}_{=\mathbf{E}A^2\mathbf{E}[B]^2 + \mathbf{E}[A]^2\mathbf{E}B^2 - \mathbf{E}[A]^2\mathbf{E}[B]^2} \\
&\quad + \underbrace{\mathbf{E}[BAB^2]}_{=\mathbf{E}A\mathbf{E}B^3} + \underbrace{\mathbf{E}[B^2A^2]}_{=\mathbf{E}A^2\mathbf{E}B^2} + \underbrace{\mathbf{E}[B^2AB]}_{=\mathbf{E}A\mathbf{E}B^3} + \underbrace{\mathbf{E}[B^3A]}_{=\mathbf{E}A\mathbf{E}B^3} + \mathbf{E}B^4 \\
&= \mathbf{E}A^4 + 4\mathbf{E}A^3\mathbf{E}B + 4\mathbf{E}A^2\mathbf{E}B^2 + 4\mathbf{E}A\mathbf{E}B^3 + \mathbf{E}B^4 \\
&\quad + 2(\mathbf{E}A^2\mathbf{E}[B]^2 + \mathbf{E}[A]^2\mathbf{E}B^2 - \mathbf{E}[A]^2\mathbf{E}[B]^2).
\end{aligned}$$

Definition 5 (Stieltjes transform). The Stieltjes transform of a free random variable is the following moment generating function:

$$G_A(z) := \frac{1}{z} + \frac{\mathbf{E}A}{z^2} + \frac{\mathbf{E}A^2}{z^3} + \cdots .$$

It is also called the Cauchy transform.

Definition 6 (R-transform). The *R-transform* is defined in terms of the inverse of the Stieltjes transform as follows:

$$R_A(w) := G_A^{-1}(w) - \frac{1}{w} = \frac{r_{A,2}}{w^2} + \frac{r_{A,3}}{w^3} + \dots .$$

Theorem 3 (addition of free random variables). *The free convolution can be expressed in terms of the R -transform as follows:*

$$R_{A+B}(z) = R_A(z) + R_B(z).$$

Example (adding two semicircles)

III.4 Poster Projects

Ideas for posters related to free probability

Why do random matrices follow free probability?

What about other operations like multiplication or truncation?

What is the Free Central Limit Theorem?

What other examples can be done exactly w/ R-transforms, eg., Marchenko–Pastur?

Do additions of random matrices follow local universality e.g. Tracy–Widom?